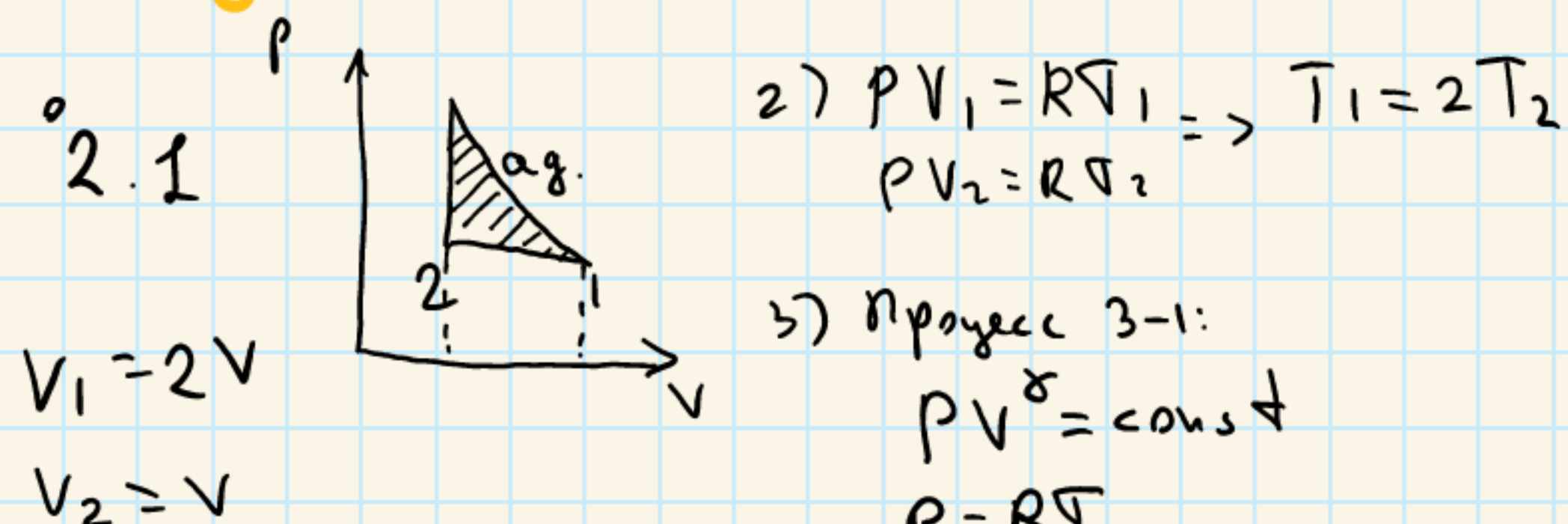


Семинар 2

12 февраля 2026 г.

16:53

Кулёвни



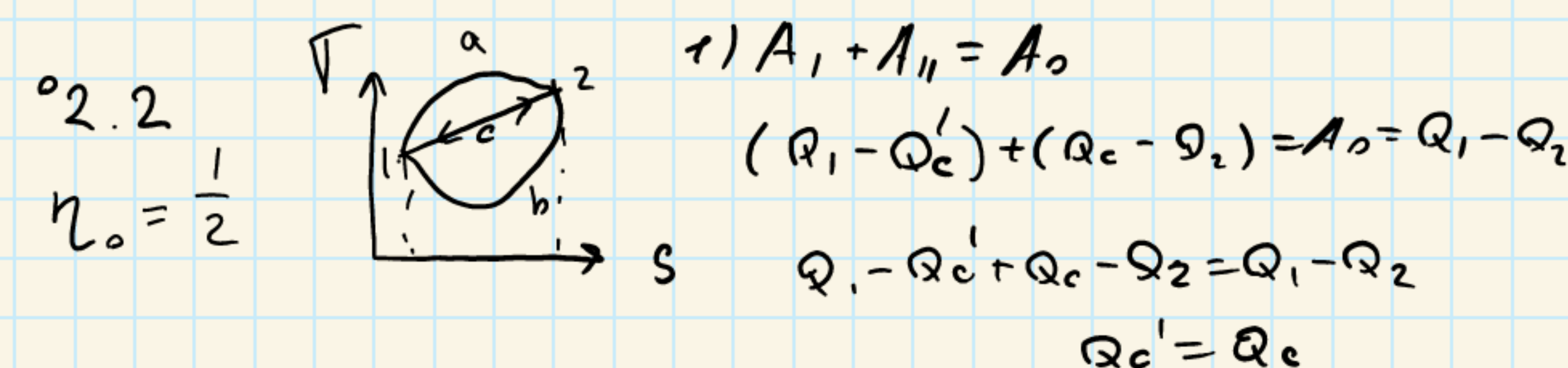
$$A_{12} = P(V_2 - V_1) = -PV_2 = -RT_2 \cdot 0,5$$

$$A_{23} = 0$$

$$A_{31} = -\Delta U = \frac{5}{2}R(T_3 - T_1) = 2,5 \cdot 0,3RT_1$$

$$Q_{12} = \frac{5}{2}R(T_3 - T_2) = \frac{5}{2}RT_1(1,3 - 0,5) = 0,8 \cdot \frac{5}{2}RT_1$$

$$\eta = \frac{-0,5RT_1 + 0,80RT_1}{2RT_1} = \frac{0,3}{2} \approx 15\%$$



$$Q_1 + Q_2 = Q_c + Q'_c = 2Q_c$$

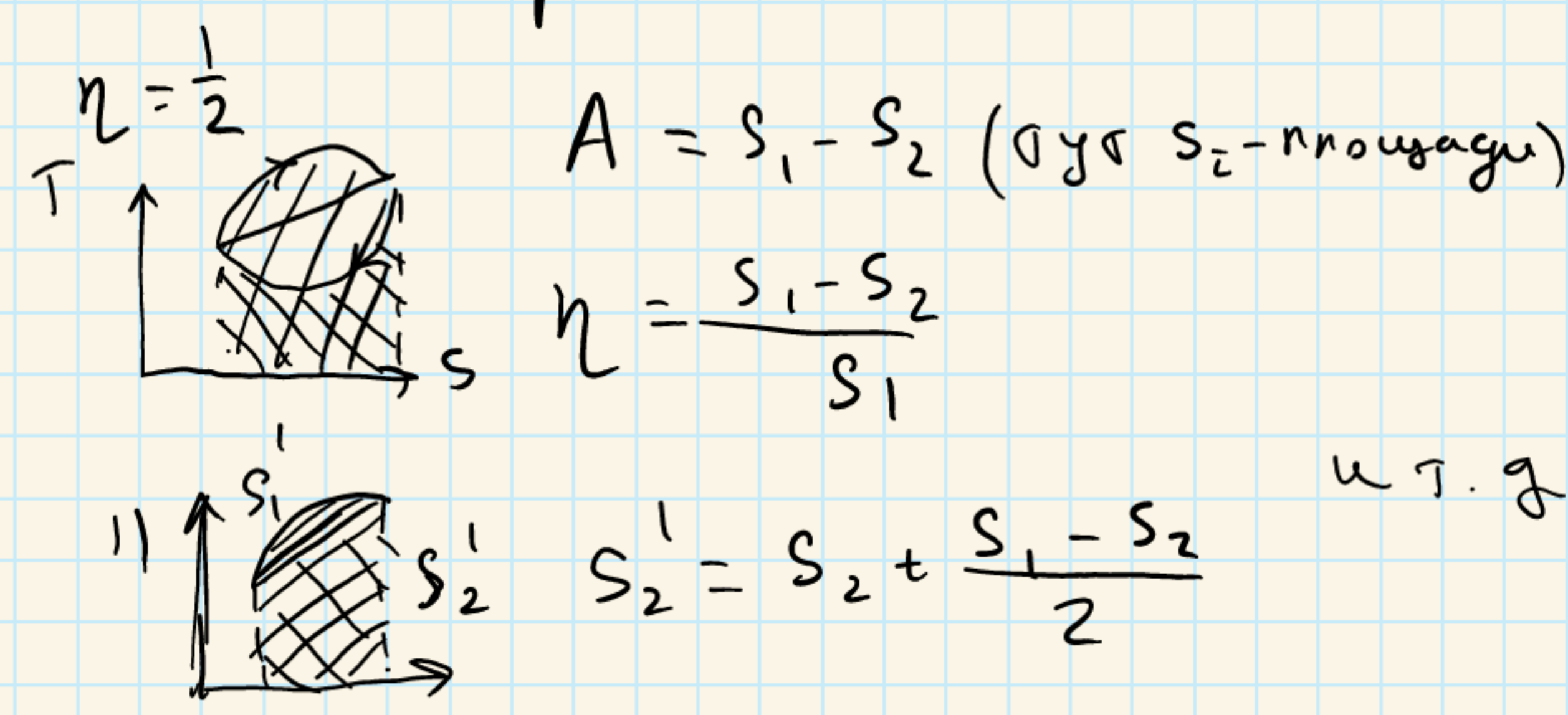
$$Q_c = \frac{Q_1 + Q_2}{2} = \frac{Q_1 + \frac{Q_1}{2}}{2} = \frac{(3/2)Q_1}{2} = \frac{3}{4}Q_1$$

$$\eta_1 = \frac{A_1}{Q_1} = \frac{1}{4}, A_1 = Q_1 - Q'_c = Q_1 - Q_c = Q_1 - \frac{3}{4}Q_1 = \frac{1}{4}Q_1$$

2) По условию $Q_c = \frac{3}{4}Q_1$, $A_{11} = Q_c - Q_2 = \frac{3}{4}Q_1 - \frac{1}{2}Q_1 = \frac{1}{4}Q_1$

$$\eta_2 = \frac{A_{11}}{Q_c} = \frac{\frac{1}{4}Q_1}{\frac{3}{4}Q_1} = \frac{1}{3}$$

Решение семинариста:



Сам семинар

№3.25

Дано

$$C = 1 \frac{\text{ккал}}{2 \cdot \text{кг}}$$

$$q = 80 \text{ ккал/кг}$$

Решение

$$\text{const} = m_1 T_1 = 373 \text{ К} \rightarrow T$$

$$\delta Q_1 = C m_1 dT$$

$$\delta Q_2 = q dm_2$$

$$m_2 T_2 = 273 = \text{const}$$

$$dU = Tds - PdV$$

$$\oint ds = 0 = \frac{\delta Q_1}{T} + \frac{\delta Q_2}{T_2} \Rightarrow \frac{T_3}{T_1} \delta Q_1 = Q_2$$

$$Q_2 = \int q dm_2 = -q m_2$$

$$Q_1 = \int_{T_1}^{m_2 T_2} \frac{T_2 C m_2 dT}{T} = T_2 C m_2 \ln \frac{T_2}{T_1} = -q m_2$$